ISOMORPHISM CLASSES OF TORSORS UNDER AN ABELIAN SHEAF ARE CLASSIFIED BY FIRST COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The leading context span is the author's own Definition definition-torsor (tag 03AH) of the same section, which fixes the meaning of pseudo torsor, torsor, morphism of torsors and trivial torsor used in the counted statement; it defines that label and is therefore not listed among the external references.

Editorial note. The author's closing sentence at proof lines 359-360, which declares the verification that the two constructions are mutually inverse to be omitted, is printed verbatim in the proof span; the card records it in the core obligations and in the difficulty reason, and nothing was supplied in its place.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex): a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros. The native class line, the xr-hyper externaldocument block and the reference/slogan/history display environments are omitted because the assembler supplies a plain amsart wrapper and the excerpt uses none of those environments; no macro was invented or redefined.

Editorial note. The locator of the single external pointer refers to a verbatim excerpt of the project's own tag list, sources/stacks-sites-cohomology/referenced-tags.txt, carrying its stable Stacks tag.

Allowed standard prerequisites (editorial).

- Cohomology of abelian sheaves on a site as a right derived functor, and the vanishing of higher cohomology of injective sheaves. Derived Categories, Stacks tag 015B (derived-lemma-higher-derived-functors), invoked at proof line 353 for H^1(C, I) = 0. Lets an arbitrary class in H^1(C, H) be represented by a global section q of the quotient Q = I/H for an embedding of H into an injective abelian sheaf I, which is what the second construction starts from.
- Sheafification of a presheaf and the free abelian sheaf on a sheaf of sets. the chapter's standing constructions, used at proof lines 300-304 and again at proof lines 586-589. Produces the free abelian sheaf Z[F] on an H-torsor F and, in the surjectivity step, the sheafification L = L_1^# of the presheaf (F(U) times O(U))/O^*(U).
- Pushout of a short exact sequence of abelian sheaves along a map, and the long exact cohomology sequence with its boundary operator. standard homological algebra of abelian sheaves on a site, used at proof lines 313-333. Builds the extension E of Z by H out of the map a : Ker(sigma) to H and then produces the class xi_F as the boundary image of 1 in H^0(C, Z).

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Source reference. Original lemma 015B: sources/stacks-sites-cohomology/referenced-tags.txt, source lines 4–4. Cross-chapter pointer cited at proof line 353 for the vanishing of H^1(C, I) on an injective abelian sheaf; the embedding, the quotient Q = I/H, the lifting of the class and the verification that the preimage is a torsor all stay inside the counted proof.

Definition 0.1. Let $\mathcal{C}$ be a site. Let $\mathcal{G}$ be a sheaf of (possibly non-commutative) groups on $\mathcal{C}$. A *pseudo torsor*, or more precisely a *pseudo $\mathcal{G}$-torsor*, is a sheaf of sets $\mathcal{F}$ on $\mathcal{C}$ endowed with an action $\mathcal{G} \times \mathcal{F} \to \mathcal{F}$ such that

(1) whenever $\mathcal{F}(U)$ is nonempty the action $\mathcal{G}(U) \times \mathcal{F}(U) \to \mathcal{F}(U)$ is simply transitive.

A *morphism of pseudo $\mathcal{G}$-torsors* $\mathcal{F} \to \mathcal{F}'$ is a morphism of sheaves of sets compatible with the $\mathcal{G}$-actions. A *torsor*, or more precisely a *$\mathcal{G}$-torsor*, is a pseudo $\mathcal{G}$-torsor such that in addition

(2) for every $U \in \mathrm{Ob}(\mathcal{C})$ there exists a covering $\{U_i \to U\}_{i \in I}$ of U such that $\mathcal{F}(U_i)$ is nonempty for all $i \in I$.

A *morphism of $\mathcal{G}$-torsors* is a morphism of pseudo $\mathcal{G}$-torsors. The *trivial $\mathcal{G}$-torsor* is the sheaf $\mathcal{G}$ endowed with the obvious left $\mathcal{G}$-action.

Lemma 0.2. *Let $\mathcal{C}$ be a site. Let $\mathcal{H}$ be an abelian sheaf on $\mathcal{C}$. There is a canonical bijection between the set of isomorphism classes of $\mathcal{H}$-torsors and $H^1(\mathcal{C}, \mathcal{H})$.*

Proof. Let $\mathcal{F}$ be a $\mathcal{H}$-torsor. Consider the free abelian sheaf $\mathbf{Z}[\mathcal{F}]$ on $\mathcal{F}$. It is the sheafification of the rule which associates to $U \in \mathrm{Ob}(\mathcal{C})$ the collection of finite formal sums $\sum n_i[s_i]$ with $n_i \in \mathbf{Z}$ and $s_i \in \mathcal{F}(U)$. There is a natural map

$$\sigma : \mathbf{Z}[\mathcal{F}] \longrightarrow \underline{\mathbf{Z}}$$

which to a local section $\sum n_i[s_i]$ associates $\sum n_i$. The kernel of σ is generated by sections of the form $[s] - [s']$. There is a canonical map $a : \mathrm{Ker}(\sigma) \to \mathcal{H}$ which maps $[s] - [s'] \mapsto h$ where h is the local section of $\mathcal{H}$ such that $h \cdot s' = s$. Consider the pushout diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{Ker}(\sigma) & \longrightarrow & \mathbf{Z}[\mathcal{F}] & \longrightarrow & \underline{\mathbf{Z}} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle a} & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{H} & \longrightarrow & \mathcal{E} & \longrightarrow & \underline{\mathbf{Z}} & \longrightarrow & 0 \end{array}$$

Here $\mathcal{E}$ is the extension obtained by pushout. From the long exact cohomology sequence associated to the lower short exact sequence we obtain an element $\xi = \xi_{\mathcal{F}} \in H^1(\mathcal{C}, \mathcal{H})$ by applying the boundary operator to $1 \in H^0(\mathcal{C}, \underline{\mathbf{Z}})$.

Conversely, given $\xi \in H^1(\mathcal{C}, \mathcal{H})$ we can associate to ξ a torsor as follows. Choose an embedding $\mathcal{H} \to \mathcal{I}$ of $\mathcal{H}$ into an injective abelian sheaf $\mathcal{I}$. We set $\mathcal{Q} = \mathcal{I}/\mathcal{H}$ so that we have a short exact sequence

$$0 \longrightarrow \mathcal{H} \longrightarrow \mathcal{I} \longrightarrow \mathcal{Q} \longrightarrow 0$$

The element ξ is the image of a global section $q \in H^0(\mathcal{C}, \mathcal{Q})$ because $H^1(\mathcal{C}, \mathcal{I}) = 0$ (see Derived Categories, Lemma 015B). Let $\mathcal{F} \subset \mathcal{I}$ be the subsheaf (of sets) of sections that map to q in the sheaf $\mathcal{Q}$. It is easy to verify that $\mathcal{F}$ is a $\mathcal{H}$-torsor.

We omit the verification that the two constructions given above are mutually inverse. $\square$